

PAPER CODE: BMC104 ROLL NO. _____

**BBS INSTITUTE OF PROFESSIONAL STUDIES
PRAYAGRAJ**

FIRST SESSIONAL EXAM (2025-2026)

COURSE: MCA SEM: 1st
SUBJECT: Discrete Mathematics Branch: MCA
MAX MARKS: 20 MAX TIME: 1:30 HR

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A)

[5 X 1 = 5]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Prove that $a+1 = 1$.
- Prove Reversal Law in Composite Relation. $(SoR)^{-1} = R^{-1}oS^{-1}$
- Prove that for any set A, B and C:
 $(A - (A \cap B)) = A - B$
- Express power set of each of these sets.
1) $\{O, \{O\}\}$ 2) $\{a, \{a\}\}$
- Show that the statements: $P \rightarrow Q$ and $\sim Q \rightarrow \sim P$ are equivalent.

(SECTION B)

[4 X 2 = 8]

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

- Convert $\sim(p \vee q) \leftrightarrow (p \wedge q)$ into DNF.
- Check if the relation $R = \{(a,a), (b,b), (c,c), (a,b), (b,c)\}$ is antisymmetric on set $A = \{a,b,c\}$.

Verify whether the following compound propositions is tautology or contradiction or Contingency $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

- Minimize the following Boolean function by using k-map
 $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$
- Let L1 be the lattice defined as D6 and L2 be the lattice $(P(S), \subseteq)$, where $P(S)$ be the power set defined on set $S = \{a, b\}$. Justify that the two lattices are isomorphic.
- Let A be the set of non-zero integers & let R be the relation on $A \times A$ defined by $(a,b) R (c,d) \leftrightarrow ad = bc$. Show that the relation is an equivalence relation.

(SECTION C)

[2X3.5= 7]

Q.3 ATTEMPT ANY TWO OF THE FOLLOWING

- Construct the Hasse Diagram for $(P(S), \subseteq)$ where $P(S)$ is a power set defined on set $S = \{a, b, c\}$. Determine whether it is a Lattice or not.
- Define Modular Lattice. Justify that if 'a' and 'b' are the elements in abounded distributive lattice and if 'a' has complement a' , then
I) $a \vee (a' \wedge b) = a \vee b$ II) $a \wedge (a' \vee b) = a \wedge b$.
- Prove that the relation R defined on the set of positive integers $(x,y) \in R$ if $x \cdot y$ is divisible by 5 is an equivalence relation.

(Following paper code and roll no. to be filled by student at the answer books)								
PAPER CODE:(BMC-101)	ROLL NO.							

(Following paper code and roll no. to be filled by student at the answer books)								
PAPER CODE:(BMC-101)	ROLL NO.							

**BBS COLLEGE OF ENGINEERING & TECHNOLOGY
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**BBS COLLEGE OF ENGINEERING & TECHNOLOGY
PRAYAGRAJ**

SEM: 1st

SUBJECT: Fundamental of Computers and Emerging Technologies

MAX TIME: 1:30 HR

NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.

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(SECTION A)

[5 X 1 = 5]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Define the term Operating System.
- Compare computer hardware and software with example.
- Write an algorithm to check whether a given number is even or odd.
- What is non-volatile memory?
- What is the difference between algorithm and pseudo code?

(SECTION B)

[4 X 2 =08]

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

- Discuss multiprocessing, multitasking and multiprocessor operating system.
- How command based is different from GUI based operating system?
- Discuss some differences between LANs and WANs.
- Write the difference between application software and system software.
- Write the pseudo code to calculate the factorial of a number.
- Differentiate among assembler, compiler and interpreter.

(SECTION C)

 $[2 \times 3.5 = 7]$

Q3. ATTEMPT ANY TWO OF THE FOLLOWING:

- Define the term algorithm. Discuss its characteristics. Write an algorithm to find maximum of three numbers.
- What is topology? Describe four basic topologies with their respective advantages and limitations.
- Build a flow chart to check whether given year is a leap year or not. Give any two inputs to show the working.

(Following paper code and roll no. to be filled by student at the answer books)

PAPER CODE: BMCT06 ROLL NO.

**BBS COLLEGE OF ENGINEERING & TECHNOLOGY
PRAYAGRAJ**

FIRST SESSIONAL EXAM (2025-2026)

COURSE: MCA

SUBJECT: Cyber Security

MAX MARKS: 20

SEM: 1st

Branch: MCA

MAX TIME: 1:30 HR

**NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.**

(SECTION A)

[5 X 1 = 5]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Explain CIA triads used for Cyber Security.
- Explain the needs of Cyber Security.
- Illustrate the concept of Information Assurance.
- What is DoS attack in Cyber Security.
- List out various threats in E-Commerce.

(SECTION B)

[4 X 2 = 08]

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

- Elaborate the role of backups and archival storage for data security consideration. (2)
- List out various Security threats in Cyber Security. explain any two. (2)
- Explain the concept of Access Control used for security. Also list out various types of Access Controls.
- Discuss in details about IDS. Differentiate between HIDS and NIDS.
- Explain the advantages of VPN? How does the system work? (2)
- Discuss the concept of Electronic Payment System and its types. (1)

(SECTION C)

[2X3.5= 7]

Q3. ATTEMPT ANY TWO OF THE FOLLOWING:

- Illustrate the concept of Information System. List out the various types of Information Systems. explain any one of them.
- Define the role and working of firewall. List out the various types of firewalls. explain any two.
- Explain the concept of Digital Signature with the help of model. Explain the concept of Public and Private Keys used in Encryption/Decryption.

(Following paper code and roll no. to be filled by student at the answer books)

PAPER CODE-(BMC-103) ROLL NO.

**BBS COLLEGE OF ENGINEERING & TECHNOLOGY
PRAYAGRAJ**

FIRST SESSIONAL EXAM (2025-2026)

COURSE: MCA

SEM: 1st

SUBJECT: PRINCIPLES OF MANAGEMENT & COMMUNICATION

MAX MARKS: 20

MAX TIME: 1:30 HR

**NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.**

(SECTION-A)

[5 X 1 = 5]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Define Management.
- Mention any two roles of manager.
- Name the founder of Scientific Management.
- Define the term planning.
- What is meant by Esprit de Corps?

(SECTION-B)

[4 X 2 =08]

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

- Is Management Art or Science? Explain.
- What are the limitations of the planning function?
- Briefly mention the importance of organising function.
- Mention the principles of Scientific Management.
- Briefly state the features of planning function.

(SECTION-C)

[2X3.5= 7]

Q3. ATTEMPT ANY TWO OF THE FOLLOWING:

- Briefly discuss the process of planning.
- Explain Henry Fayol's 14 principles of Management.
- Discuss the various functions of Management.

(Following paper code and roll no. to be filled by student at the answer books.)	
PAPER CODE:-(BMC-102)	ROLL NO. 25MCA0008

**BBS COLLEGE OF ENGINEERING & TECHNOLOGY
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FIRST SESSIONAL EXAM (2025-2026)**

COURSE: MCA **SEM: 1st**
SUBJECT: Problem Solving Using C
MAX MARKS: 20 **MAX TIME: 1:30 HR**

**NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.**

(SECTION A) **[5 X 1 = 5]**

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- Write an algorithm to print sum of numbers from 1 to 10.
- Describe the rules of declaring identifiers.
- What do you mean by actual and formal parameters in function?
- What happens if you create a loop that never ends?
- Predict how long the following loop runs?
for (int x = 1; x = 3; x++)

(SECTION B) **[4 X 2 = 08]**

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

- Write a recursive function to find factorial of a number.
- What do you mean by precedence and associativity of operators? Explain with a suitable example.
- Differentiate between for, while and do-while loops with suitable example.
- What do you mean by call by value and call by reference? Explain working with proper syntax.
- Explain the following with proper example:
(i) break (ii) continue (iii) goto
- Write a program to check the given number is palindrome or not.

(SECTION C) **[2X3.5= 7]**

Q3. ATTEMPT ANY TWO OF THE FOLLOWING:

- Write the syntax of switch statement. Explain with suitable example. Write a program to print number of days in a month by using switch statement. Ex: Input=12 Output= 31 Days. Input=2 Output=28/29 Days.
- Write a program to swap the content of two variables that reflect both call by value and call by reference effect in main and called function.
- Discuss about the following operators in C with example.
(i) Bitwise operators (ii) Logical operators (iii) sizeof () operator

(Following paper code and roll no. to be filled by student at the answer books)									
PAPER CODE: BMC105	ROLL NO.								

**BBS COLLEGE OF ENGINEERING & TECHNOLOGY
PRAYAGRAJ**

FIRST SESSIONAL EXAM (2025-2026)

COURSE: MCA

SEM: I

SUBJECT: COA

Branch- MCA

MAX MARKS: 20

MAX TIME: 1:30 HR

- NOTE: 1. QUESTION PAPER CONTAINS THREE PARTS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A)

[5 X 1 = 5]

Q1. ATTEMPT ALL FROM THE FOLLOWING:

- What is Program Counter?
- Define Addressing Mode.
- What are the different types of instruction formats?
- What is GPR?
- Define the term computer architecture.

(SECTION B)

[4 X 2 =08]

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

- What is register transfer language?
- Explain in detail the principle of carry looks ahead adder and design 4-bit CLA adder.
- What is bus arbitration?
- Represent the following decimal number in IEEE standard floating-point format in a single & double precision method representation method. (i) $(1243.725)_{10}$
- Explain IEEE-754 standard for floating point representation.
- Discuss the role of ALU in a processor?

(SECTION C)

[2X3.5= 7]

Q3. ATTEMPT ANY TWO OF THE FOLLOWING:

- What are the different types of Buses used in computer architecture?
- Show the multiplication process using Booth's algorithm when the following numbers are multiplied: $7 * 3$
- What do you mean by Processor Organization?